

KUSHAGRA MAHAJAN

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EDUCATION

Carnegie Mellon University, School of Computer Science Master of Computational Data Science — GPA: 4.05/4.00 Coursework: Machine Learning, Machine Learning for Large Datasets, Natural Language Processing, Multilingual NLP, Multimodal ML, Cloud Computing, Visual Learning and Recognition, Best Capstone Solution Award	Pittsburgh, PA Aug. 2021 – Dec. 2022
Indraprastha Institute of Information Technology Delhi (IIIT Delhi) Bachelor of Technology in Computer Science — GPA: 8.78/10.00 Coursework: Machine Learning (Teaching Assistant - Fall '18), Computer Vision, Probabilistic Graphical Models, Data Mining, Collaborative Filtering, Data Structures and Algorithms, Compiler Design, Database Systems, Operating Systems.	Delhi, India Aug. 2014 – Dec. 2018

EXPERIENCE

Amazon.com Software Development Engineer — Amazon Alexa Building proactive, personalized experiences and natural language subscriptions for Alexa Notifications.	Seattle, WA June 2025 – Present
Amazon Web Services Software Dev Engineer — AWS Bedrock Part of the Model Hosting and Optimizations team. Built scalable, fault-tolerant, and efficient infrastructure for training, inference, and fine-tuning of some of the largest foundation models from leading AI companies.	Seattle, WA Aug. 2023 – June 2025
Carnegie Mellon University Research Associate — Advisor: Prof. Louis-Philippe Morency - Collaborated with Meta AI to conduct research on multimodal ML based user satisfaction estimation for VR/AR glasses (Quest Pro), leveraging 3D head/body pose, normalized view generation, IMU, facial expressions, language, acoustic, and physiological signals; applied multimodal representation learning techniques like HighMMT, and Neural Mixed Effect Models. - Built and deployed AI systems for navigation, mapping, interaction of a embodied agent virtual robot based on user-assigned real-world tasks in collaboration with Amazon Alexa AI.	Pittsburgh, PA Feb. 2023 – Aug. 2023
TCS Research and Innovation Labs Research Engineer — Computer Vision Team - Designed a meta-learning based computer vision framework for skin and chest disease diagnosis, focusing on medical image classification and segmentation; published 3 papers [2, 3, 4] and built and deployed products across several hospitals. - Developed a fast, memory-efficient end-to-end document image alignment and information extraction system using a novel keypoint extraction algorithm; published 1 paper [5] and 1 granted US patent [1]. - Investigated techniques for disentangling biological signal from noise in cellular images, applying CNNs with ArcFace loss and distributed training, resulting in top-5 percentile ranking in the NeurIPS 2019 challenge (96.06% accuracy).	Delhi, India Jan. 2019 – April 2021
Intel Corporation Machine Learning Engineer Intern Advisor: Mr. Tigi Thomas Created a highly optimized sensor-based hand gesture detection and recognition model used for gesture controls for on-device deployment surpassing benchmarks for memory constraints and output latency, tested on laptop hardware.	Bangalore, India Aug. 2017 – Dec. 2017

PROJECTS

Multimodal Image to Recipe Generation Course Project: Multimodal ML Advisor: Prof. Louis-Philippe Morency Generative AI transformer based recipe generation from food images using novel approach comprising co-learning, component-aware embeddings, contrastive loss for semantic similarity, and improved evaluation metrics. [Report]	Aug. 2022 – Nov. 2022
Natural Language Inference for Code-Switched Hinglish Course Project: Multilingual NLP Advisor: Prof. Alan Black Improved state-of-the-art NLI performance on GLUECoS benchmark by 6% for code-switched Hindi-English by translating to matrix or embedded language, and domain adaptation of multilingual LLM transformer models XLM-R, mBERT, mT5 etc. [Report]	Jan. 2022 – April 2022
Twitter Cloud Native Web Service with Microservices Course Project: Cloud Computing Advisor: Prof. Majd Sakr Built an ETL pipeline using Spark for processing 1.2TB Twitter data, and a microservice based architecture using Kubernetes for data retrieval and running analytic jobs in a cost constrained setting.	Jan. 2022 – April 2022

PATENTS AND PUBLICATIONS

[1] **K. Mahajan**, M. Sharma, L. Vig, Tata Consultancy Services Limited. “Method and System for Keypoint Extraction from Images of Documents”. [US Patent US12136286B2](#) [\[Link\]](#)

[2] **K. Mahajan**, M. Sharma, L. Vig. “Meta-DermDiagnosis: Few-Shot Skin Disease Identification using Meta-Learning”. [IEEE International Conference on Computer Vision and Pattern Recognition 2020 Workshops \(CVPRW\)](#). [\[PDF\]](#)

[3] **K. Mahajan**, T. Khurana, A. Chopra, I. Gupta, C. Arora, A. Rai. “Pose Aware Fine-Grained Visual Classification Using Pose Experts”. [25th IEEE International Conference on Image Processing \(ICIP\) 2018](#). [\[PDF\]](#)

[4] T. Khurana, **K. Mahajan**, C. Arora, A. Rai. “Exploiting Texture Cues for Clothing Parsing in Fashion Images”. [25th IEEE International Conference on Image Processing \(ICIP\) 2018](#). [\[PDF\]](#)

[5] A. Pandit, **K. Mahajan**, S. Kunde. et. al. “Data-Efficient Training of High-Resolution Images in Medical Domain”. [29th European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning \(ESANN\) 2021](#). [\[PDF\]](#)

[6] **K. Mahajan**, M. Sharma, L. Vig. et. al. “CovidDiagnosis: Deep Diagnosis of COVID-19 Patients Using Chest X-Rays”. [IEEE International Workshop on Thoracic Image Analysis, MICCAI 2020](#). [\[PDF\]](#)

[7] **K. Mahajan**, M. Sharma, L. Vig. “Character Keypoint-based Homography Estimation in Scanned Documents for Efficient Information Extraction”. [CBDAR workshop at the 15th IEEE International Conference on Document Analysis and Recognition \(ICDAR\) 2019](#). [\[PDF\]](#)

SKILLS & ACHIEVEMENTS

Programming Languages, FrameWorks and Tools: Python, C, C++, Java, SQL, Tensorflow, Pytorch, Keras, Spark, PySpark, Caffe, OpenCV, AWS, Azure, Kubernetes, Scikit, NumPy, Pandas, SciPy, EspNet, Kafka, Samza, HBase, MongoDB, Docker, Git
Achievements: Best Capstone Solution Award (Masters program at CMU), *Travel Grant:* AICTE-INAE for ICIP 2018, *Dean’s List:* 2017-2018.